Mini Project: Image Enhancement for Poor Quality Images

Introduction

This project focuses on applying various image processing techniques to enhance poor quality images. Different types of degradation such as noise, blur, and low contrast are addressed using suitable methods. The goal is to improve image quality and extract meaningful information from each image.

1. Component Extraction — Circle

Technique Used

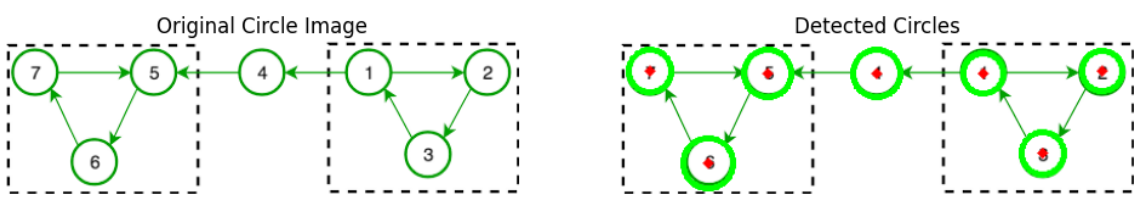
Hough Circle Transform

Why this technique was selected

This technique is effective for detecting circular shapes based on edge information.

Images

Original and processed Image:



Discussion of Results

The method successfully detected circular components with accurate centers and radii. Some parameter tuning was required to avoid false detections.

2. Component Extraction — COVID-19 Chart

Technique Used

Thresholding and Contour Detection

Why this technique was selected

Used to separate objects and detect connected components.

Images

Original and processed Image:

Original COVID-19 Chart

Final Extracted Components

Discussion of Results

Main components were successfully extracted. Some minor components may not be detected, but overall extraction is effective.

3. Enhancement of Blurry Images — Building

Technique Used

Unsharp Masking

Why this technique was selected

Enhances edges to improve clarity.

Images

Original and processed Image:

Original Building

Enhanced Building

Discussion of Results

Edges became clearer, although some blur remains due to loss of detail in the original image.

4. Enhancement of Blurry Images — Dog

Technique Used

Unsharp Masking

Why this technique was selected

Improves visibility of blurred subjects.

Images

Original and processed Image:

Original Dog

Enhanced Dog

Discussion of Results

The subject became more visible, but strong blur limits full restoration.

5. Noise Removal — Text

Technique Used

Bilateral Filtering and OTSU Thresholding

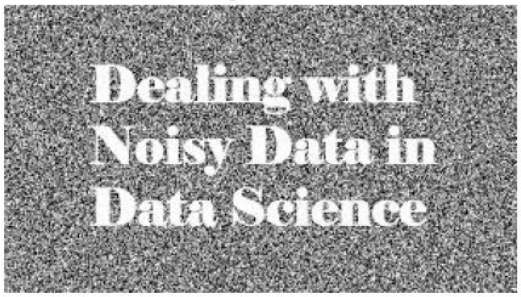
Why this technique was selected

Reduces noise while preserving edges and improves readability.

Images

Original and processed Image:

Original Text



Final Clean Text

Discussion of Results

Text readability improved significantly, although slight distortion remains.

6. Noise Removal — Rocket

Technique Used

Non-Local Means Denoising

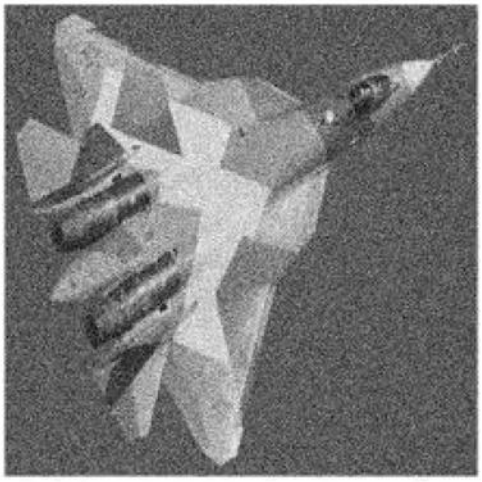
Why this technique was selected

Preserves details while removing noise.

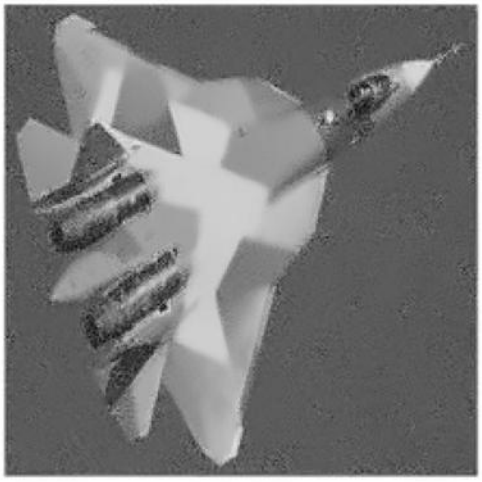
Images

Original and processed Image:

Original Rocket



Denoised Rocket



Discussion of Results

Noise was reduced effectively while maintaining important features.

7. Noise Removal — Wind Chart

Technique Used

Median Filtering and Morphological Processing

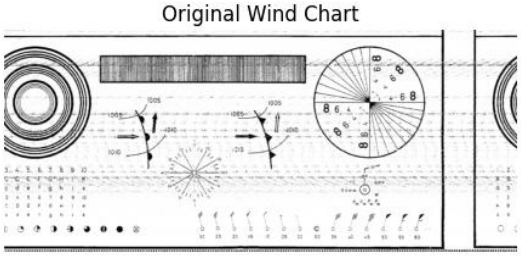
Why this technique was selected

Targets structured vertical noise.

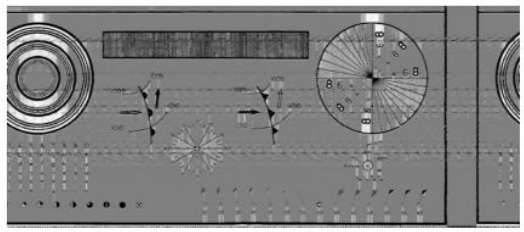
Images

Original and processed Image:

Original Wind Chart



Processed Wind Chart



Discussion of Results

Noise was reduced while preserving the chart structure, though slight fading occurred.

8. Visual Enhancement — Newspaper

Technique Used

Bilateral Filtering and OTSU Thresholding

Why this technique was selected

Improves contrast and readability.

Images

Original and processed Image:

Original Newspaper

Enhanced Newspaper

Discussion of Results

Text became clearer and more readable, though some details were lost.

9. Visual Enhancement — Name Plate

Technique Used

CLAHE

Why this technique was selected

Enhances local contrast for better readability.

Images

Original and processed Image:

Original Name Plate

Enhanced Name Plate

Discussion of Results

Text visibility improved, though some noise may be amplified.

Conclusion

Different techniques were applied based on image degradation type. The results show that selecting the appropriate method is essential for effective enhancement. Future work could include advanced restoration techniques such as deep learning.